

BTCS501N/ BTCS501

B.Tech./ B.Tech.+M.B.A./ B.Tech.+M.Tech.(CSE, IT, CSE-CF, CSE-BDA, CSE-BDCE, CSE-ES, CSE-MA, CSE-BDAI, CSE-CMCI, CSE-AII, CSE-DSI, CSE-FSDI)

V/VI Semester Examination, June-July 2024

Theory of Computation

Choice Based Credit System (CBCS)

Time: 3 Hrs.

Maximum Marks: 60

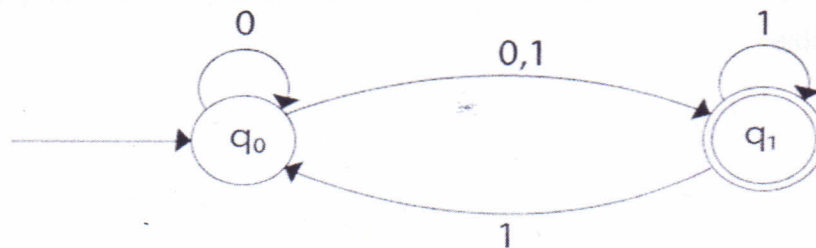
Minimum Pass Marks: 24

- Note: (1) All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.
 (2) From each question, part 'A' and 'B' are compulsory and part 'C' has an internal choice.
 (3) Draw a neat diagram, wherever necessary.
 (4) Assume suitable data wherever necessary.

Q.1(A) Differentiate between DFA & NFA with tuples and example. 03

(B) Construct an NFA with $\Sigma = \{0, 1\}$ for the language $L = \{0^m 1^n \mid m \geq 0 \text{ and } n \geq 1\}$. 03

(C) Convert the given NFA into its equivalent DFA. 06



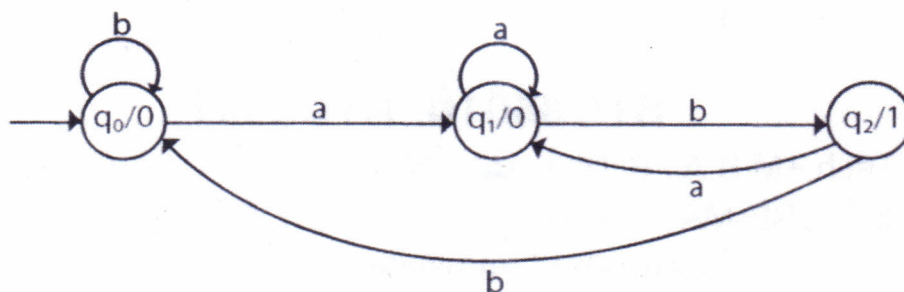
OR

Construct a NFA accepting all string in $\{a,b\}$ with either two consecutive a's or two consecutive b's, and convert to DFA Using Myhill-Nerode Theorem.

Q.2(A) What is Pumping Lemma for regular-languages? Write Application of Pumping Lemma. 03

(B) What are regular expressions? Write the regular expression for the language starting with a but not having consecutive b's. 03

Contd.....



OR

Design the following -

- I. Design a NFA from given regular expression $10 + (0 + 11)0^* 1$.
- II. Design a NFA from given regular expression $1 (1^* 01^* 01^*)^*$.

Q.3(A) How to generate a string from the grammar using leftmost derivation and right most derivation. Explain with suitable example. 03

(B) Explain in brief Closure and Decision Properties of CFLs. 03

(C) Describe Chomsky Normal Form and Greibach Normal Form with suitable example. 06

OR

Generate a Chomsky Normal Form (CNF) for the following context free grammar (CFG).

$S \rightarrow aAa|bBb|e$

$A \rightarrow C|a$

$B \rightarrow C|b$

$C \rightarrow CDE|e$

$D \rightarrow A|B|ab$

Q.4(A) Define Pushdown Automata. Also describe Instantaneous Description of PDA. 03

(B) Explain pushdown automata acceptance by empty stack with suitable example. 03

(C) Design a Pushdown Automata for the given language $L = \{a^n b^{n+m} c^m \mid n, m \geq 1\}$. 06

OR

Design a Pushdown Automata for the given language $L = \{a^n b^{2n} \mid n \geq 1\}$.

Q.5(A) What is the difference between a recursive and recursively enumerable languages? 03

(B) What is the halting problem of Turing machine and what do you mean by saying that the halting problem of TM is undecidable? 03

(C) Design a Turing machine to the language $L = \{a^n b^n\}$ where $n \geq 1$. 06

OR

Design a Turing machine which performs concatenation operation over two unary input strings.

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